Substation to Storage

A feasibility thesis (rev. 2)

Repurposing decommissioned Czech transformer stations (transformovny / rozvodny) as grid batteries with rooftop solar — instead of demolishing them

Quantitative model on realised Czech market data · reviewed & revised June 2026

Verdict — a conditional yes, for the right sites

Reusing a decommissioned medium- or high-voltage substation's grid connection for a battery (plus rooftop PV on the surrounding buildings) adds EUR 2.6m–EUR 3.2m of net present value over an equivalent greenfield battery in our representative 20 MW / 40 MWh case — from avoided connection capex, roughly two years' earlier operation, and a small avoided-demolition credit. In the base case that edge is decisive, not incremental: reuse is a viable 12%-IRR project while the identical greenfield barely breaks even at an 8% discount rate (EUR -0.1m NPV). The advantage survives even if balancing-service revenue goes to zero. It applies only where the retained connection is at least a few MW at distribution or transmission voltage; the thousands of small 0.4/22 kV kiosk trafostanice are far too small. And because the connection belongs to the grid operator, the realistic developer is the DSO/TSO, the site owner, or a partner holding a connection agreement — exactly who is already doing it at Chvaletice, Kladno and Vitkovice.

Scope & method: a representative-site techno-economic model, not site due diligence. Day-ahead arbitrage is computed from realised 2019-2026 Czech auction prices (the same LP battery backtest used elsewhere in this project); balancing revenue, capex and the connection saving are parameterised from 2025-26 Czech market data with explicit scenarios and sensitivities. Rev. 2 incorporates an adversarial fact-check and model review: capacity is now allocated between markets rather than double-counted, greenfield capex is phased, and ancillary rates are calibrated to 2026 auctions. All figures are illustrative and not investment advice.

1 The premise and the question

Across Czechia, ageing transformer and switching stations (*transformovny* and *rozvodny*) are reaching the end of their life — major substation assets (transformers, switchgear) carry design lives of roughly 30-45 years, so stations undergo major refurbishment or replacement on that cycle, and older or redundant ones are demolished outright. The question posed here: rather than demolish, could such a site host a grid battery — its cables and connection are already there — with solar panels on the surrounding buildings, and is that worth doing?

The short answer is that the buried value of these sites is almost never the building or the old iron; it is the **grid connection**. Whether reuse beats demolition is therefore a question about one number: how many megawatts, at what voltage, the connection can still carry.

2 What is actually being demolished — and which sites matter

'Transformer station' spans four orders of magnitude of capacity. The feasibility verdict flips across that range, so the taxonomy is the heart of the analysis.

Tier	Typical site	Voltage / size	Battery it supports	Verdict
A	Kiosk / pole trafostanice	0.4/22 kV, <1 MVA	< 0.5 MW	No — too small
B	Distribution substation	22-110 kV, 10-40 MVA	5-20 MW	Yes — sweet spot
C	Transmission node / ex-thermal switchyard	110-400 kV, 100s MW	50-300 MW	Ideal

The thousands of small kiosk *trafostanice* (Tier A) being swapped out in network upgrades are the wrong target: a battery there would be a fraction of a megawatt, useful at most for a local community microgrid. The prize is Tier B and C — and at the top of the range sit retired coal and industrial sites whose switchyards already carry hundreds of megawatts. One caveat: the common Czech pattern is to rebuild the substation in place — at Trebic-Ptacov the 1984 building was demolished in February 2026 while its modernised replacement ran alongside — so the connection often stays in network use. 'Free connection' is real only where genuine spare capacity remains.

3 Why the grid connection is the prize

A grid connection is the longest-lead, scarcest and often costliest part of a battery project. In Ember's benchmark all-in BESS capex of roughly $125/kWh outside China, about $50/kWh — some 40% — is installation and connection, not cells. Germany's four transmission operators say battery connection requests (over 500 GW by late 2025) far exceed available grid capacity, and are moving to maturity-based allocation from April 2026 to unclog the queue; the UK's TMO4+ reform (approved April 2025) is rebuilding a ~770 GW queue around 'ready and needed' criteria, with batteries the largest technology in it. Across Europe developers pivot to co-location precisely to share an existing connection. A live connection means skipping a two-to-four-year queue and saving on the order of the German BKZ connection charges — typically EUR 40-180 per kW, higher at congested nodes — plus the line and bay works on top.

> **The lever, in one line**
>
> A retired substation hands a battery developer the two things money cannot quickly buy: a connection agreement that already exists, and the years that would otherwise be spent waiting for one.

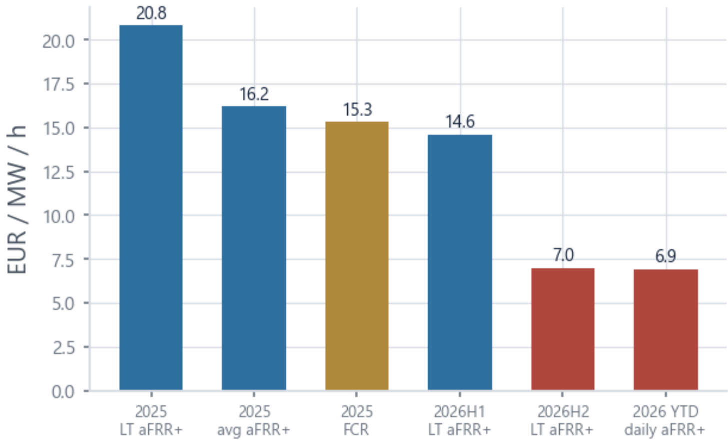
4 The Czech market right now

The timing is favourable on capacity and policy, and treacherous on revenue.

Boom and policy. Czech installed battery capacity is about 2.3 GWh (mid-2026, dominated by home systems); Aurora Energy Research expects roughly 6 GWh of *additional utility-scale* capacity by 2030 — a total approaching

8+ GWh by the end of the decade. The 2025 Energy Act amendment (Lex OZE III, zakon c. 87/2025 Sb.) created a storage licence and let standalone batteries connect and trade on organised markets from 1 October 2025; the provision allowing them to sell balancing services (FCR/aFRR/mFRR) to CEPS takes effect 1 August 2026 — which any new-build's commissioning date comfortably clears. A companion 2025 amendment streamlined grid connection and introduced deposits against speculative capacity hoarding; the Modernisation Fund (RES+ 5/2025, CZK 2bn at up to 20% of capex) co-funds standalone storage. Tellingly, the flagship projects all sit on reused industrial or coal connections: the 230 MWh Chvaletice battery at Sev.en's lignite power station (delivered April 2026), the 45 MW / 90 MWh Kladno sister project, BESS Lipnice (40 MW / 120 MWh, SUAS + EIF) in the Sokolov coal district, and CEZ ESCO's 10 MW Vitkovice battery — the country's largest when it entered service in February 2024, since overtaken several times. The thesis is not hypothetical — it is the prevailing build pattern.

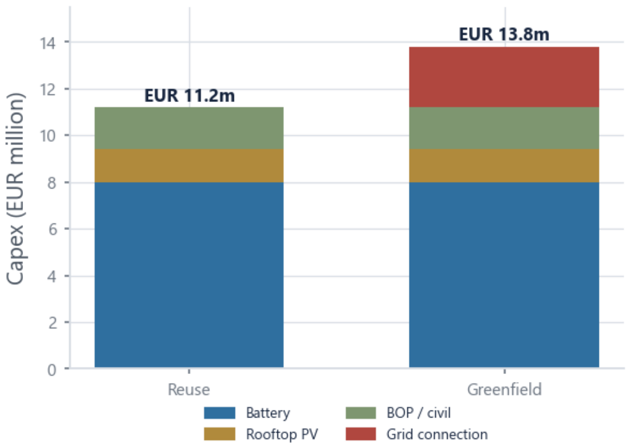
The revenue trap. But the easy money is disappearing. The battery-relevant balancing segments are small — market commentary puts aFRR at ~300 MW and FCR at ~70 MW of typically procured capacity (CEPS's official dimensioning is ~960 MW of positive reserves overall including mFRR, with only ~150 MW as a fixed aFRR floor) — and the growing fleet is flooding them. Average aFRR+ prices fell about 34% in 2025 to ~16 EUR/MW/h (long-term contracts averaged 20.8), FCR fell ~40% to ~15, and 2026 is worse: the H1 long-term aFRR+ contract cleared at 14.6, H2 at ~7.0, and the daily market averaged 6.9 EUR/MW/h through May. Czech commentators call it 'the end of the illusion of easy returns.' Nano Energies reports its optimised multi-country battery portfolio averaged EUR 340k/MW of revenue in 2025 (company-reported, not Czech-only, not profit) — a level the saturation makes hard to repeat. The durable floor is energy arbitrage — and here Czechia is unusually rich: day-ahead spreads alone were worth about EUR 88k per MW in 2025 in our LP backtest, before intraday.

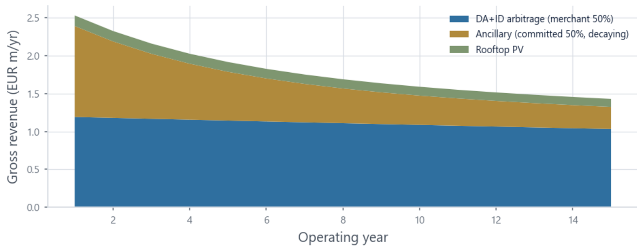


Czech CEPS balancing-service prices, EUR/MW/h — the collapse that pushes battery economics toward arbitrage. Sources: EIFlexi (Jan 2026), oEnergetice (June 2026).

5 A quantitative feasibility model

We model a representative Tier B site: a retired 110/22 kV substation with about 20 MW of usable retained connection, fitted with a 20 MW / 40 MWh (2-hour) battery and 2 MWp of rooftop PV. Crucially, capacity is **allocated**, not double-counted: a megawatt committed to FCR/aFRR cannot simultaneously run the merchant cycling schedule, so arbitrage is earned only on the uncommitted fraction (50% in the base case) and ancillary only on the committed one. Arbitrage uses the realised-price LP result with a 93% realisation factor (the DA auction clears all hours at once, so that leg is nearly implementable; the haircut mostly covers intraday execution). Ancillary starts at 2026-calibrated rates (EUR 120k per committed MW-year; 2025 realised ~40% higher) and decays toward a saturation floor. Greenfield pays 15% of capex up front, the deflated remainder at energisation, and waits 2 years in the connection queue.



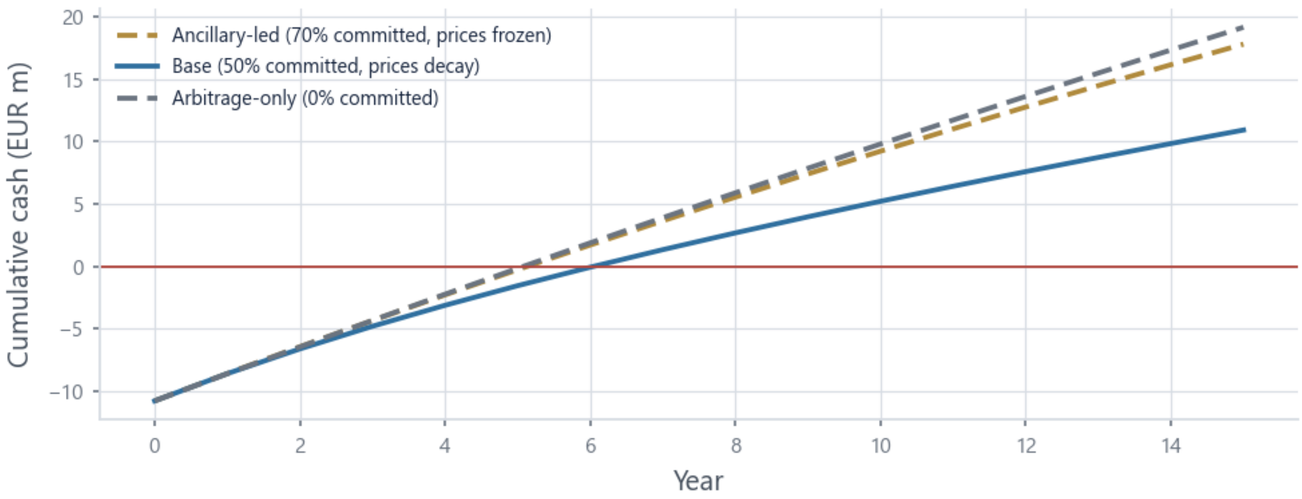


Left: capex — reuse avoids the grid-connection block (the EUR 0.4m avoided-demolition credit is netted against the spend at t=0, owner-=-developer assumption). Right: base-case gross revenue with balancing decaying into an arbitrage-and-solar floor.

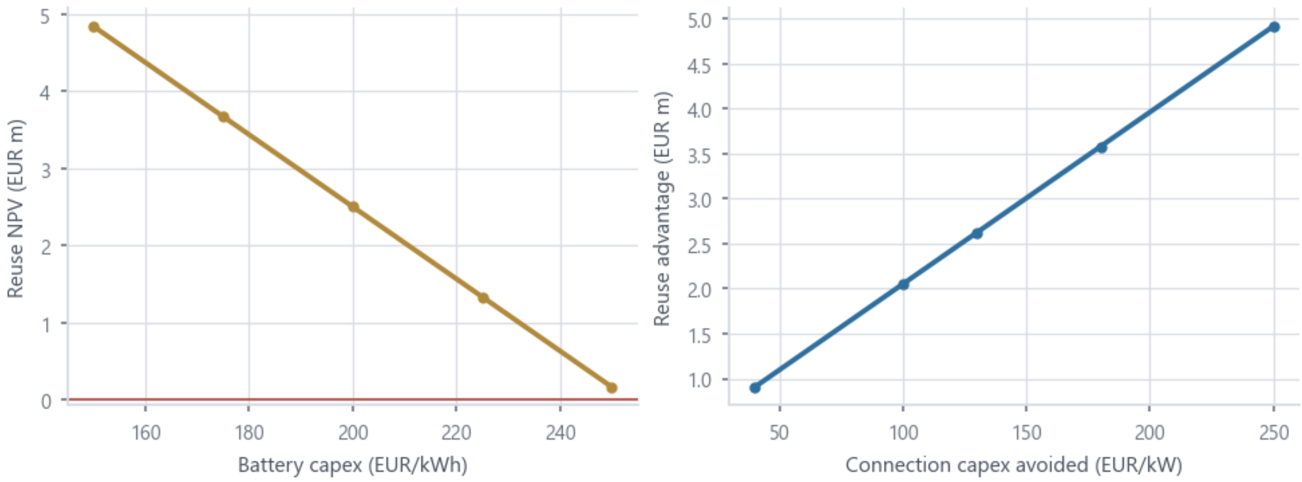
Results — reuse / greenfield, 15-year, 8% discount, simple payback.

Scenario	NPV EURm	IRR	Payback	Reuse edge EURm
Ancillary-led (70% committed, prices frozen)	6.0 / 2.9	17% / 12%	6y / 9y	+3.1
Base (50% committed, prices decay)	2.5 / -0.1	12% / 8%	7y / 10y	+2.6
Arbitrage-only (0% committed)	6.6 / 3.4	17% / 13%	6y / 9y	+3.2

Three readings. **First**, the reuse edge (EUR 2.6m–EUR 3.2m) is stable across scenarios because it is a capex-and-timing difference, not a revenue bet. **Second**, in the base case it is decisive: greenfield is roughly NPV-zero at an 8% discount rate while reuse earns 12% — the connection is the difference between a project and no project. **Third**, arbitrage-only now *beats* the base case: at 2026 auction prices, committing half the plant to balancing destroys value relative to pure merchant operation — the quantitative echo of 'the end of easy returns' (a real operator would re-allocate dynamically, so the base case is mildly conservative).



Cumulative cash for the reuse case across scenarios. Without the EUR 0.4m demolition credit, base-case simple payback is 7 years (vs 7 with it).



Left: reuse NPV vs battery capex — the project needs turnkey costs near current benchmarks to clear the 8% hurdle. Right: the reuse advantage scales almost linearly with the connection capex avoided (German BKZ range 40-250 EUR/kW shown).

Reconciliation with market benchmarks

Capstone DC estimates 5-7% unlevered IRRs for standalone French merchant BESS in 2025 — so why does reuse earn 12% here? Three auditable reasons: Czech day-ahead spreads are far richer than French ones (EUR 88k/MW/yr DA-only in 2025, from realised prices); the reuse case pays no connection capex and starts two years earlier; and our turnkey capex (EUR 200/kWh) reflects 2026 pricing. The greenfield column (8% IRR base) sits close to the European merchant range — which is exactly the point: the site, not the strategy, is the alpha.

6 Rooftop solar on the surrounding buildings

Solar is a sensible companion but a junior partner. Roof area on a substation's ancillary buildings is limited, so a couple of MWp is realistic — perhaps EUR 0.1-0.2m a year of generation at a cannibalised midday capture price. Its real value is synergy and zero marginal connection cost: the PV charges the battery cheaply at the midday solar trough (often near or below zero — Czechia saw a sharp rise in negative-price hours), which the battery then shifts into the evening peak, and both assets share the one connection and one set of balance-of-plant. It improves the economics at the margin; it does not carry the project.

7 When it is worth doing — the decision rule

- **Connection test.** Reuse beats demolition when the retained connection carries at least roughly 5 MW at 22 kV or above and real spare capacity exists. Below that, the connection is worth less than the cleared land — demolish.
- **Owner test.** The connection is the DSO's or TSO's asset. The credible developer is the grid operator, the site owner, or a partner with a connection agreement — not an outside party assuming the cables are free for the taking. (The avoided-demolition credit likewise belongs to the site owner and counts only when owner = developer.)
- **Condition test.** The value is the line, the bay, the land and the permits — not the old transformer, which a battery replaces with its own inverters and transformer anyway. Brownfield liabilities (contamination, asbestos, foundations) must net out below the connection saving.
- **Counter-incentive.** Demolition itself can be subsidised — MMR programmes fund demolition and brownfield revitalisation, though mainly for municipalities and public entities — and the operator may need the bay for the replacement substation. Reuse must clear that bar too.

8 Risks

- **Balancing saturation.** Modelled explicitly — and already biting: 2026 long-term aFRR+ cleared ~65% below the 2025 average. The base case commits only half the plant and decays the price; the arbitrage-only scenario is the stress case.
- **Arbitrage cannibalisation.** The same battery build-out that saturates balancing also narrows the day-ahead spreads arbitrage relies on — so even the arbitrage-only case is optimistic over a full 15 years; treat it as an upper bound on the floor.
- **Bankability.** Capstone DC (Nov 2025) puts standalone French merchant BESS at 5-7% unlevered IRR, some markets ~3%; Czech spreads are richer but the revenue is exposed, not contracted. Our greenfield base case (~8% IRR) is consistent with that picture.
- **Brownfield and technical.** Contamination, protection upgrades, connection re-rating and aFRR energy-reserve requirements for a 2-hour system can erode the saving; due diligence per site is essential.

9 Verdict

Worth doing — selectively, and as the grid owner

For Tier B and C sites, reusing the connection for a battery beats both greenfield and demolition. The connection is the scarce asset, and reuse banks its value plus two years of lead time — EUR 2.6m of NPV in the base case, an edge that holds even if balancing revenue disappears, and one that in today's market is the difference between a financeable project and a marginal one. It is not a strategy for the small kiosk stations, and it is realistically executed by the grid operator or a connection-holding partner — precisely the pattern playing out across the Czech coal and industrial fleet. The revenue numbers will keep falling as the market saturates; the connection-reuse advantage will not.

References

- CzechTrade / ess-news — Czech large-scale storage; AlphaESS Chvaletice 230 MWh & Kladno 45 MW/90 MWh (Seven Batteries, final delivery Apr 2026); SUAS+EIF BESS Lipnice 120 MWh; CEZ ESCO Vitkovice 10 MW (in service Feb 2024).
- EIFlexi (Jan 2026) & oEnergetice (Jun 2026) — CEPS balancing auction results 2025-26 ('konec iluze snadnych vynosu'); aFRR+/FCR price series; CEPS SVR dimensioning ~960 MW positive reserves.
- Zakon c. 87/2025 Sb. (Lex OZE III) — storage licence, market access from 1 Oct 2025, balancing services to CEPS from 1 Aug 2026.

• Aurora Energy Research (Nov 2025) — ~+6 GWh utility BESS in CZ by 2030. Solarni novinky (Apr 2026) — Nano Energies EUR 340k/MW (2025, company-reported, multi-country).

• Ember, 'How cheap is battery storage?' (Dec 2025) — $125/kWh all-in ex-China/US, ~40% installation+connection. BNEF (Dec 2025) — Europe turnkey avg $177/kWh. interconnector.de (2025) — German BKZ 40-180 EUR/kW.

• Capstone DC (Nov 2025) — 5-7% unlevered IRR, French merchant BESS. Modo Energy — intraday uplift 35-96% over DA-only (GB).

• German TSOs (2025) — >500 GW battery connection requests, maturity-based allocation from Apr 2026. Ofgem/NESO TMO4+ (Apr 2025) & DESNZ/Ofgem letter (Apr 2026) — UK queue reform.

• Trebicsky denik (Feb 2026) — rozvodna Trebic-Ptacov rebuilt in place. MMR — demolition/brownfield programmes (mainly public applicants).

• This project — CZ day-ahead LP arbitrage backtest on realised 2019-2026 EPEX prices (Energy-Charts / Fraunhofer ISE).

Disclaimer. Research and educational material only. Backtests use realised data with stated, simplified cost and execution assumptions and do not constitute investment advice. Past performance does not indicate future results.

Appendix The feasibility model (code)

The full techno-economic model used above (rev. 2: capacity allocation, realisation factor, phased greenfield capex, demolition credit as a t=0 offset). Assumptions are named constants with sources in comments; the scenarios and sensitivity sweeps are the output.

python · analysis/substation_battery.py

```python
"""
Substation-to-storage feasibility model (v2 — post-review).

Thesis: a decommissioned MV/HV transformer substation ("transforma■ní /
rozvodná stanice") slated for demolition retains the one thing a new
battery project waits years and pays the most for — a live grid
connection at meaningful capacity. Reusing it for a battery (with
rooftop PV on the surrounding buildings) can beat both a greenfield
battery and plain demolition. This model quantifies that for a
representative site and stress-tests the result.

Revenue is built bottom-up, with CAPACITY ALLOCATED between markets —
a MW committed to FCR/aFRR capacity cannot simultaneously run the full
merchant cycling schedule, so the two streams are earned on disjoint
fractions of the connection (this replaces the additive stack of v1,
which double-counted capacity and produced upside-case IRRs):

  - Day-ahead arbitrage on the merchant fraction: from the LP battery
    backtest on realised CZ auction prices (results/battery_daily_CZ.csv),
    times a documented intraday uplift, times a realisation factor
    (the DA leg of the LP is nearly implementable because the auction
    clears all hours at once; the haircut mostly covers the ID leg).
  - Ancillary services on the committed fraction: a per-committed-MW
    rate calibrated to 2026 Czech auction levels (2025 realised ~40%
    higher, but H2-2026 long-term aFRR+ already cleared at ~7 EUR/MW/h),
    decaying toward a floor as the ~GW-scale Czech balancing market
    saturates. Note: batteries may sell balancing services to CEPS from
    1 Aug 2026 (Lex OZE III), which any new-build COD comfortably clears.
  - Rooftop PV on surrounding buildings: secondary; shares the
    connection and charges the battery through the midday trough.

The reuse case differs from greenfield by (a) avoided grid-connection
capex, (b) ~2 years earlier commercial operation, and (c) a small
avoided-demolition credit (valid only where site owner = developer).
Greenfield capex is PHASED (development spend up front, equipment near
COD, with deflation on the delayed spend) so the timing advantage is
not overstated.

Every assumption is a named constant with its source in the comment;
the three scenarios and the sensitivity sweep are the real output. The
verdict is conditional, not a single number.
"""

from __future__ import annotations

import json
import sys
from dataclasses import dataclass, asdict
from pathlib import Path

import numpy as np
import pandas as pd

sys.path.insert(0, str(Path(__file__).parent.parent))

RES = Path(__file__).parent.parent / "results"
OUT = RES


# -------------------------------------------------------------- economics --
@dataclass
class SiteConfig:
    # --- representative "Tier B" site: a retired 110/22 kV distribution
    #     substation with ~20 MW of usable retained connection capacity ---
    power_mw: float = 20.0
    duration_h: float = 2.0          # 2-hour system -> 40 MWh
    pv_mwp: float = 2.0              # rooftop PV on surrounding buildings

    # --- capex (EUR) ---
    capex_batt_eur_per_kwh: float = 200.0   # turnkey LFP 2026 (BNEF EU avg ~$177/kWh; DE >10MW <250)
    capex_pv_eur_per_kwp: float = 700.0     # CZ commercial rooftop
    capex_bop_eur_per_kw: float = 90.0      # integration/civil at a reused site
    # the lever: greenfield grid connection avoided by reuse (EUR per kW);
    # German BKZ benchmarks ~40-180 EUR/kW, congested nodes higher
    connection_capex_eur_per_kw: float = 130.0
    demolition_avoided_eur: float = 400_000.0  # t=0 capex offset; owner=developer only
    dev_capex_fraction: float = 0.15        # greenfield: development spend at t=0
    capex_deflation_pa: float = 0.05        # deflation on greenfield's delayed equipment spend
```

```
    # --- revenue: capacity allocation between markets ---
    # fraction of power committed to FCR/aFRR capacity products; the
    # merchant (arbitrage) schedule runs only on the remainder
    ancillary_fraction_base: float = 0.5
    ancillary_fraction_rich: float = 0.7

    # --- year-1 revenue rates ---
    arb_da_eur_per_mw: float = 88_400.0     # CZ 2025 DA-only LP (data-grounded)
    intraday_uplift: float = 0.45           # Modo: ID adds ~35-96% over DA-only
    arb_realization: float = 0.93           # blended: ~0.96 on DA leg (auction clears
                                            # all hours at once), ~0.88 on the ID leg
    ancillary_eur_per_mw_y1: float = 120_000.0  # per COMMITTED MW, 2026-calibrated
                                            # (2025 realised ~170k; H1-26 LT aFRR+ 14.6,
                                            #  H2-26 ~7.0 EUR/MW/h -> decline is live)
    ancillary_floor_eur_per_mw: float = 35_000.0
    ancillary_decay: float = 0.20           # per year toward the floor

    # --- PV revenue ---
    pv_yield_mwh_per_mwp: float = 1_000.0   # CZ ~950-1050
    pv_capture_eur_per_mwh: float = 70.0    # cannibalised solar-hour capture

    # --- opex & lifetime ---
    opex_pct_of_capex: float = 0.02
    reserved_capacity_eur_per_mw_yr: float = 6_000.0  # ongoing DSO charge proxy
    revenue_degradation: float = 0.02       # per-year derate (fade/augmentation)
    arb_growth: float = 0.01                # mild structural widening then flat
    life_years: int = 15
    discount_rate: float = 0.08
    greenfield_delay_years: int = 2         # connection queue avoided by reuse

    @property
    def energy_mwh(self) -> float:
        return self.power_mw * self.duration_h


# scenario -> (ancillary-committed fraction, ancillary price path)
def _scenario_params(cfg: SiteConfig, scenario: str) -> tuple[float, str]:
    if scenario == "ancillary_rich":
        return cfg.ancillary_fraction_rich, "frozen"
    if scenario == "arbitrage_only":
        return 0.0, "zero"
    return cfg.ancillary_fraction_base, "decay"


def _capex(cfg: SiteConfig, reuse: bool) -> dict:
    batt = cfg.energy_mwh * 1000 * cfg.capex_batt_eur_per_kwh
    pv = cfg.pv_mwp * 1000 * cfg.capex_pv_eur_per_kwp
    bop = cfg.power_mw * 1000 * cfg.capex_bop_eur_per_kw
    conn = cfg.power_mw * 1000 * cfg.connection_capex_eur_per_kw
    total = batt + pv + bop + (0.0 if reuse else conn)
    return {"battery": batt, "pv": pv, "bop": bop,
            "connection": (0.0 if reuse else conn), "total": total}


def _annual_revenue(cfg: SiteConfig, year: int, scenario: str) -> dict:
    """Year-indexed revenue (year 0 = first operating year).

    Arbitrage runs on the merchant fraction of power only; ancillary is
    paid per committed MW on the committed fraction. Components are all
    degraded identically so they sum exactly to gross.
    """
    as_frac, anc_path = _scenario_params(cfg, scenario)
    deg = (1 - cfg.revenue_degradation) ** year

    arb_unit = (cfg.arb_da_eur_per_mw * (1 + cfg.intraday_uplift)
                * cfg.arb_realization * (1 + cfg.arb_growth) ** year)
    arb = arb_unit * cfg.power_mw * (1 - as_frac) * deg

    if anc_path == "frozen":
        anc_unit = cfg.ancillary_eur_per_mw_y1
    elif anc_path == "zero":
        anc_unit = 0.0
    else:  # decay toward the saturation floor
        anc_unit = (cfg.ancillary_floor_eur_per_mw
                    + (cfg.ancillary_eur_per_mw_y1 - cfg.ancillary_floor_eur_per_mw)
                    * (1 - cfg.ancillary_decay) ** year)
    anc = anc_unit * cfg.power_mw * as_frac * deg

    pv = cfg.pv_mwp * cfg.pv_yield_mwh_per_mwp * cfg.pv_capture_eur_per_mwh * deg
    return {"arbitrage": arb, "ancillary": anc, "solar": pv,
            "gross": arb + anc + pv}


def _npv(rate: float, cashflows: list[float]) -> float:
    return float(sum(cf / (1 + rate) ** t for t, cf in enumerate(cashflows)))
```

```python
def _irr(cashflows: list[float]) -> float | None:
    lo, hi = -0.9, 2.0
    f_lo, f_hi = _npv(lo, cashflows), _npv(hi, cashflows)
    if f_lo * f_hi > 0:
        return None
    for _ in range(200):
        mid = (lo + hi) / 2
        f = _npv(mid, cashflows)
        if abs(f) < 1.0:
            return mid
        if f_lo * f > 0:
            lo, f_lo = mid, f
        else:
            hi = mid
    return (lo + hi) / 2


def cashflow(cfg: SiteConfig, scenario: str, reuse: bool,
             demolition_credit: bool = True) -> pd.DataFrame:
    """Reuse: full capex at t=0 (minus the demolition offset), operating
    from year 1. Greenfield: development spend at t=0, deflated equipment
    spend at t=delay, operating from year delay+1. Both get exactly
    life_years operating years. Opex runs in operating years only."""
    cap = _capex(cfg, reuse)
    delay = 0 if reuse else cfg.greenfield_delay_years
    opex = cap["total"] * cfg.opex_pct_of_capex + \
        cfg.reserved_capacity_eur_per_mw_yr * cfg.power_mw
    horizon = cfg.life_years + delay
    rows = []
    for t in range(horizon + 1):
        rev = {"gross": 0.0, "arbitrage": 0.0, "ancillary": 0.0, "solar": 0.0}
        if t == 0:
            if reuse:
                net = -cap["total"]
                if demolition_credit:
                    net += cfg.demolition_avoided_eur
            else:
                net = -cap["total"] * cfg.dev_capex_fraction
        elif not reuse and t == delay:
            net = -(cap["total"] * (1 - cfg.dev_capex_fraction)
                    * (1 - cfg.capex_deflation_pa) ** delay)
        elif t <= delay:
            net = 0.0
        else:
            rev = _annual_revenue(cfg, t - delay - 1, scenario)
            net = rev["gross"] - opex
        rows.append({"year": t, "net": net, **rev})
    return pd.DataFrame(rows).set_index("year")


def evaluate(cfg: SiteConfig, scenario: str) -> dict:
    out = {}
    for reuse in (True, False):
        df = cashflow(cfg, scenario, reuse)
        cfs = df["net"].tolist()
        cum = np.cumsum(cfs)
        payback = next((i for i, v in enumerate(cum) if v >= 0), None)
        cap = _capex(cfg, reuse)
        op = df[df["gross"] > 0]
        entry = {
            "capex_total": round(cap["total"]),
            "capex_connection": round(cap["connection"]),
            "npv": round(_npv(cfg.discount_rate, cfs)),
            "irr": (lambda r: round(r, 4) if r is not None else None)(_irr(cfs)),
            "payback_years": payback,  # simple (undiscounted) payback
            "first_op_year_gross": round(op["gross"].iloc[0]) if len(op) else 0,
        }
        if reuse:
            # payback is knife-edge on the demolition credit — report both
            cfs_nc = cashflow(cfg, scenario, True, demolition_credit=False)["net"].tolist()
            cum_nc = np.cumsum(cfs_nc)
            entry["payback_years_ex_credit"] = next(
                (i for i, v in enumerate(cum_nc) if v >= 0), None)
        out["reuse" if reuse else "greenfield"] = entry
    out["reuse_advantage_npv"] = round(out["reuse"]["npv"] - out["greenfield"]["npv"])
    return out


def sensitivity(cfg: SiteConfig) -> dict:
    """Sweeps report BOTH the reuse NPV and the reuse-vs-greenfield
    advantage — the connection-capex parameter only moves the latter
    (reuse pays no connection capex by construction)."""
    grid = {}
    sweeps = {
        "capex_batt_eur_per_kwh": [150, 175, 200, 225, 250],
        "ancillary_decay": [0.05, 0.10, 0.20, 0.30, 0.40],
        "ancillary_fraction_base": [0.3, 0.4, 0.5, 0.6, 0.7],
        "connection_capex_eur_per_kw": [40, 100, 130, 180, 250],
```

```
        "discount_rate": [0.06, 0.08, 0.10, 0.12],
    }
    for field_name, vals in sweeps.items():
        grid[field_name] = []
        for v in vals:
            c = SiteConfig(**{**asdict(cfg), field_name: v})
            ev = evaluate(c, "base")
            grid[field_name].append({"value": v,
                                     "npv_reuse": ev["reuse"]["npv"],
                                     "advantage": ev["reuse_advantage_npv"]})
    return {"base_npv": evaluate(cfg, "base")["reuse"]["npv"], "sweeps": grid}


def load_arb_from_backtest() -> float | None:
    """Use the realised CZ 2025 DA-only LP revenue if available."""
    fp = RES / "battery_daily_CZ.csv"
    if not fp.exists():
        return None
    df = pd.read_csv(fp, index_col=0)
    df.index = pd.to_datetime(df.index, utc=True)
    y2025 = df[df.index.year == 2025]
    return float(y2025.revenue.sum()) if len(y2025) else None
```